Pronob Kumar Barman

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EDUCATION

University of Maryland, Baltimore County (UMBC) — Expected Dec 2025
Ph.D. Information Systems | GPA: 3.85/4.00
Research Focus: Human-Centered Generative Models & Recommendation Systems

University of Kentucky — May 2021
M.S. Statistics | GPA: 3.85/4.00
Thesis: Statistical Approaches for Healthcare Data Analytics

University of Dhaka — Jan 2017
M.S. Statistics | GPA: 3.46/4.00
B.S. Statistics | GPA: 3.46/4.00

PROFESSIONAL EXPERIENCE

PhD Researcher | UMBC — Jan 2022–Present

- Pioneered human-centered AI research at the healthcare-ML intersection, developing fair and interpretable models that demonstrably improved patient outcomes in online health communities.
- Led end-to-end development of **Generative Peer Support Recommender**, a groundbreaking framework that bridges machine learning with social science theories to autonomously create personalized support groups. Core technical innovations:
  - Architected **Group-specific Dirichlet Multinomial Regression (gDMR)**: Advanced DMR modeling by implementing novel group-specific parameters that integrate BERT embeddings with user demographics and multi-dimensional interaction patterns (Python).
  - Engineered **Group-specific Structural Topic Model (gSTM)**: Achieved 60% improvement in topic model stability through innovative integration of group-aware topic distributions with network embedding techniques (R).
- Validated approach through rigorous real-world experiments, demonstrating **75% higher recommendation accuracy** compared to state-of-the-art baseline models using comprehensive A/B testing with clinically relevant patient engagement metrics.
- Established research impact by open-sourcing production-quality implementations: Published gSTM R package and gDMR Python codebase, enabling widespread adoption and scalable deployment across healthcare AI research domains.

Teaching Assistant | UMBC — Jan 2022–Present

- Designed and delivered specialized workshops on **advanced ML algorithms**, **LLM fine-tuning workflows**, and **production system reliability**, receiving average student satisfaction scores of 4.8/5.0.
- Mentored 30+ graduate students in implementing and deploying scalable ML pipelines on **AWS SageMaker** and **Google Vertex AI**, resulting in 12 successful industry-ready projects.
- Developed comprehensive tutorial materials for **RAG systems** and **MLOps practices** now incorporated into the department's core graduate curriculum.

Machine Learning Engineer | Technuf LLC — Jun 2024–Aug 2024

- Architected and deployed enterprise-grade **retrieval-augmented generation (RAG)** pipelines tailored for cybersecurity threat detection and mitigation, utilizing **vector databases**, **dense retrieval via BERT embeddings**, and **OpenAI's GPT models** to transform unstructured SIEM security logs into actionable and context-aware threat intelligence.
- Developed a rigorous evaluation framework utilizing **qualitative and quantitative assessment techniques**, including manual review and validation by cybersecurity analysts, significantly enhancing the reliability and accuracy of generated threat intelligence.
- Engineered an advanced anomaly detection system employing statistical models and deep learning-based methods, **reducing 20% false-positive alerts** and achieving notable improvements in threat response efficiency, saving approximately 10 hours of analyst effort weekly.

Machine Learning Engineer | Technuf LLC — Aug 2022–Dec 2022

- Optimized critical data infrastructure by implementing advanced **SQL query optimization** techniques for **200K+ healthcare records**, reducing processing latency by **40%** and enabling real-time analytics capabilities.
- Designed and deployed comprehensive **Power BI visualization suite** with interactive dashboards that transformed raw healthcare metrics into actionable business intelligence, adopted by executive leadership for strategic decision-making.
- Implemented automated CI/CD testing pipelines with **Flutter**, reducing QA cycles by 30% and accelerating product release velocity while maintaining quality standards.

Deputy Director, Data Science | Bangladesh Bank — Apr 2018–Dec 2021

- Led development of machine learning-based credit risk assessment models, deploying predictive algorithms that reduced loan defaults by **10% across national banking portfolio** (approximately $25M in prevented losses).
- Spearheaded econometric analysis initiative for foreign direct investment, implementing data-driven strategy recommendations that increased **FDI inflows by 15%** and strengthened key economic sectors.

- Supervised team of **5 data scientists** in building and maintaining mission-critical financial intelligence dashboards used by central bank executives for monetary policy decisions.

TECHNICAL SKILLS

- **Programming & Languages:** Python, R, Java, SQL, MATLAB, SAS
- **ML & AI Frameworks:** PyTorch, TensorFlow, scikit-learn, Keras
- **Generative AI:** GPT-3/4, LangChain, Hugging Face, BERT
- **NLP & Text Mining:** SpaCy, NLTK, Transformers, Word2Vec
- **Cloud & Deployment:** AWS SageMaker, GCP, Docker, Kubernetes
- **Data Management:** PostgreSQL, Pinecone, ChromaDB
- **Visualization:** Tableau, Power BI, Matplotlib, Seaborn
- **Research Tools:** NVivo, Qualtrics, LaTeX, Git, VSCode

SELECTED PROJECTS

DistilBERT-based Question Answering System GitHub

- Deployed **DistilBERT** models on AWS SageMaker for real-time customer **query resolution**.
- Reduced response time by **30%** through scalable API integration.

Deploying a Machine Learning Model for Digit Classification on Google Cloud Run Github

- Developed and deployed a scalable **Flask-based web** app using **TensorFlow**, containerized with **Docker**, and hosted on **Google Cloud Run** to enable real-time predictions through a public API endpoint.

Air Quality Prediction and Deployment on Google Cloud Platform Github

- Designed and deployed a scalable **Air Quality Index (AQI)** forecasting system using linear regression, containerized with **Docker**, and integrated with a **Flask** web app on **Google Cloud Platform** for interactive user predictions.

LLM Generative AI with LangChain and OpenAI API Github

- Developed a **retrieval-augmented generation system** for advanced semantic search, using **Pinecone and ChromaDB**.

End-to-End Medical Chatbot with Llama2 and Hugging Face API Github

- Developed a **medical chatbot with Llama2**, improving **patient interaction flow by 30%** through **LangChain** enhancements.

Cognitive and Affective State Recognition in iVR Using Multimodal Signals Github

- Developed inverse inference models using **multimodal signals**, integrating neuro-physiological data such as EEG and eye-tracking to map cognitive and affective states in immersive virtual reality (iVR).

Predictive Policing with GAN

- Implemented a **Generative Adversarial Network (GAN)** model to identify and mitigate bias in predictive policing algorithms, achieving a 20% reduction in false-positive rates.

PUBLICATIONS

- **Barman, P. K.**, Reynolds, T. L., & Foulds, J. (2025). "Facilitating Online Healthcare Support Group Formation using Topic Modeling." *Proceedings of the 19th World Congress on Medical and Health Informatics (MedInfo 2025)*, 2025.
- Rachael, K., Reynolds, T. L., Yus R. & **Barman, P. K.** (2024). "Acceptance of Occupancy Dashboards among Undergraduate Students: Towards Everyday Decision-making Infrastructure that can be Leveraged in Pandemics." *ACM Conference on Computer-Supported Cooperative Work and Social Computing*, 2024. (Under Review)
- Walid, M. A. A., Devnath, M. K., Muiz, B., Melon, M. M. H., **Barman, P. K.**, & Barman, P. K. (2024). "Efficient Graph-Based Intrusion Detection for CAN Bus Security with Minimal Training Data." *Proceedings of the 6th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI 2024)*, 2024.
- Balogun, Z. A., **Barman, P. K.**, Onwumbiko, B. K., & Reynolds, T. L. (2024). "User-generated Data for Different Types of Mobile-Personal Health Records: A Computationally-guided Qualitative Analysis Approach." *AMIA Annual Symposium, 2024*.
- Faruq, M. O., Walid, M. A. A., Baowaly, M. K., Devnath, M. K., Ejaz, M. S., **Barman, P. K.**, & Sattar, A. S. (2024). "Shielding Privacy: A Technique of Extenuating Composition Attacks in Various Independent Data Publication." *Bulletin of Electrical Engineering and Informatics, 13(4), 2677-2686*.
- **Barman, P. K.** (2023). "Detecting Abnormal Health Conditions in Smart Home Using a Drone." *arXiv preprint arXiv:2310.05012*.

GRANTS & AWARDS

UMBC GSA Research Grant (2024): $1,000 for research on healthcare group formation models.